

Public Finance:

Economic Impact Analysis of Government



Edoardo Cefala



Module Info



A maximum of 100 points can be achieved in total.

- **3 homework assignments, 6 points each = 18 points**
- **2 tests at 25 points each -> April 24 and June 12, 2026**
- **Quizzes= 10 Points**
- **Participation** in the course units = **10 points**
- **Report** on a selected research paper in Focus Area 2 = **12 points**

Minimum Criterion: For a positive overall result, a total of 25 points (50%) must be achieved on the test.

A retake of the exam is possible if 15 points were achieved.

- **Grading Scale**
 - ≥ 90 points "Very Good"
 - ≥ 80 points "Good"
 - ≥ 70 points "Satisfactory"
 - ≥ 60 points "Sufficient" (Pass)

- **Lecturers:**
 - Edoardo Cefala, edoardo.cefala@wu.ac.at

- Lecture slides
- Papers for the reports
- Additional materials on Canvas

- The structure of the course is based on "Public Finance and Public Policy" by Jonathan Gruber.

- Excerpts from:
 - Stiglitz, Rosengard (2015) "Economics of the Public Sector"
 - Hindriks, Myles (2nd Edition) "Intermediate Public Economics"

... .on a paper of your choice!

Formatting Criteria

- PDF document, your name in the file name & header!
- 1-2 pages
- English
- Coherent phrasing, a clear central theme, correct spelling, citations
- No cover page, table of contents...

Submission

On Canvas, once submitted, it is final.

The submission will be checked for plagiarism! If the text does not pass this check, you will receive 0 points, and the assignment cannot be retaken!

Simply translating sentences from the paper into German also counts as plagiarism!

Content Structure

Part 1 - Summary of the paper: A clear and correct description of what the paper is about, Research question, objectives, methods, results,... (max. 1 page!)

- **Part 2:** Discuss which aspect of the paper was most interesting to you. What was particularly well done in the paper? Why are you convinced of this?
- **Part 3:** What is missing in the paper? Describe what you think is underdeveloped in the paper, why this aspect would be important, and how you would improve/supplement it. Stick to the context of the present paper and its objective – do not pose a new research question! Refer to the course content and previous texts.

Try to work as independently as possible: formulate results, methodology, and the research question in your own words! Dare to question and critique the papers. Our main goal is for you to engage with the papers critically and independently!

- And please, not with ChatGPT! (We run your reports through an AI detector.)
- **Why?** Because we want you to engage with the papers independently, and AI-based applications analyze complex studies inadequately.
- Furthermore, we discuss the papers in class, and you will benefit the most if you have read them yourself!
- Last but not least, it's relevant for the exam – so you're saving yourself time by doing it yourself.

1. LV

- Commodity Taxation: Model of tax incidence, empirical examples.

2. LV

- Inefficiencies of taxation in the model
- Externalities, CO2 tax, carbon inequality.

3. LV

- Homework and paper discussion.

4. LV

- Gender in Public Finance: Taxes and gender, labor market.

5. LV

- Taxes and transfers in the labor market model – Aggregate.

6. LV

- **Individual** incentives with taxes: Optimal taxation, income and substitution effect.

7. LV

- Homework and paper discussion.

8. LV

- Incentives with taxes and **transfers**: Model, example of basic income, unemployment, and the special case of information asymmetry.

9. LV

- Taxation of the factor capital: Intertemporal decision model, capital in Austria.

10. LV

- Homework and paper discussion.

Papers



Tips & Tricks for reading papers on
Canvas

Paper 1: Can VAT Cuts and Anti-Profiteering Measures Dampen the Effects of Food Price Inflation?

Benzarti et al (2024): *"Can VAT Cuts and Anti-Profiteering Measures Dampen the Effects of Food Price Inflation?"* NBER Working Paper No. 32241. R&R American Economic Review.

Setting Argentina

- 21% VAT reduction from August to December 2019 on selected basic food items
- Anti-profiteering measures after the tax cut was lifted.
- Dynamic Difference-in-Difference
- **Data:** Retail scanner data, consumer survey.

Results

- Asymmetrical pass-through when no anti-profiteering measures exist; otherwise symmetrical (i.e., equally strong for VAT increases & decreases)
- Anti-profiteering measures prevent regressive effects
- Households with higher incomes benefit more from the measure.

Paper 2: Investing in infants: The lasting effects of cash transfers to new families.

- Barr, A., Eggleston, J., & Smith, A. A. (2022). *The Quarterly Journal of Economics*, 137(4), 2539-2583.

Research Question: What are the long-term effects of cash transfers in early childhood on educational and labor market outcomes?

- Regression Discontinuity Design (RDD)
- Special focus on income and education.
- Long-term cost-benefit calculation!
- Which designs lead to the greatest benefit?
- Efficient design of redistribution measures

Paper 3: You've got mail: A randomised field experiment on tax evasion

- Bott et al. (2020). *Management science*, 66(7), 2801-2819.

Research Question : How do moral appeals and the perceived probability of detection influence the behavior of taxpayers?

- **Randomized field experiment** with 15,000 taxpayers in Norway.
- Participants had a high probability of having incorrectly reported foreign income.
- Treatment: Letters sent to taxpayers with varying content :
 - **Moral appeals** (importance of paying taxes) or increasing the perceived probability of detection (probability of being caught for tax evasion).
- **Result:** Moral appeals increase tax compliance, showing high savings potential

Introduction – Public Finance, what is that?



Chapter 1

What is Public Finance?

"Public finance is the economic doctrine/theory of the public sector and its activities. It encompasses the analysis of the effect of administrative regulations as well as public revenues and expenditures."

- "Public finance studies the role of the state in the economy."

What is Public Finance?

- **Positive economics** deals with how the world is: A factual point of discussion that can be demonstrated empirically and scientifically." → "An increase in the value-added tax leads to a rise in consumer prices."
- **Normative economics** deals with how the world should be: It therefore involves value judgments and is not "refutable" – its statements are based on ethical and political beliefs. → "The government should reduce income inequality through higher taxes for the rich."
- Public finance is an interface between positive and normative economics. It uses scientific methods to model or evaluate the effects of political measures and assesses them with regard to societal values such as efficiency, justice, or sustainability

The 4 Questions of Public Finance

- **When** should the government intervene?
- **How** can the government intervene?
- **What** are the effects of government interventions?
- **Why** do governments implement specific measures?

When should the government intervene?

- **Market Failure:** Whenever the allocation achieved by the market is not the most efficient one:
 - Positive/negative externalities (e.g., without government intervention, under-provision of health insurance, education. High emission of greenhouse gases).
 - Public goods (e.g., national defense, infrastructure).
 - Asymmetric information (e.g., health insurance). Imperfect competition (e.g., monopoly formation).
 - Individual behavioral failures (e.g., pension savings, alcohol).
- **Redistribution** When the allocation achieved by the market is not the one desired by society -> values, fairness, equality, poverty, etc..

How can the government intervene?

- **Taxes or Subsidies**

Price mechanism: Consumption/allocation is to be influenced by "artificially" increasing or decreasing the price.

- **Restrict or Mandate Consumption**

e.g., youth protection laws or compulsory insurance.

- **Public Provision / Subsidies for Private Provision**

e.g., infrastructure, education, private research institutes.

- **Redistribution Measures**

Through taxes and transfer payments: e.g., wealth taxes, minimum income support, etc..

What are the **effects** of government interventions?

Direct Effects

Are those effects we would **expect** if individuals **do not change** their behavior due to the interventions.

Example: The introduction of wealth taxes increases tax revenues by exactly the predicted amount (no side effects like tax avoidance/evasion or emigration are considered).

Indirect Effects (Behavioral)

Effects that arise in **addition** to direct effects because individuals **adapt their behavior** to the intervention.

Example: After the introduction of a wealth tax, particularly wealthy individuals emigrate -> tax revenues decrease.

Empirical focus of Public Economics/Finance

Goal: Identification of causal relationships.

- Event Studies
- Regression Discontinuity
- Synthetic Control
- Difference-in-Differences (DiD)

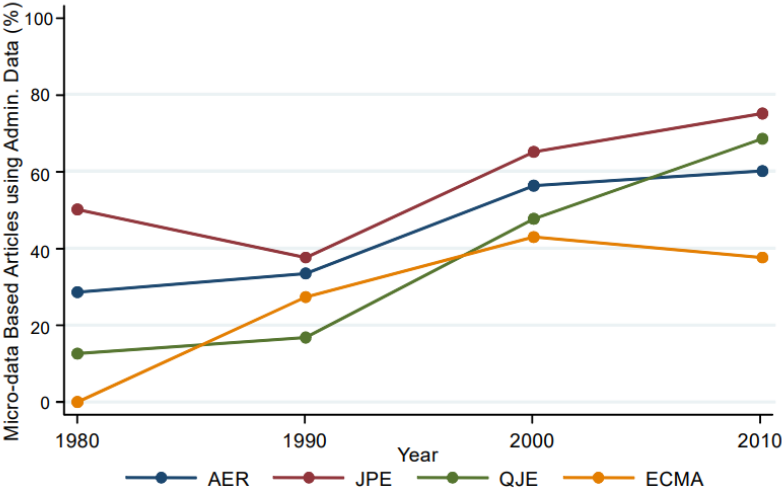
Large datasets are needed for quasi-experimental methods:

- Tax and social security data (register data).
- School district databases.
- Scanner data (purchases)

Advantages of register data?

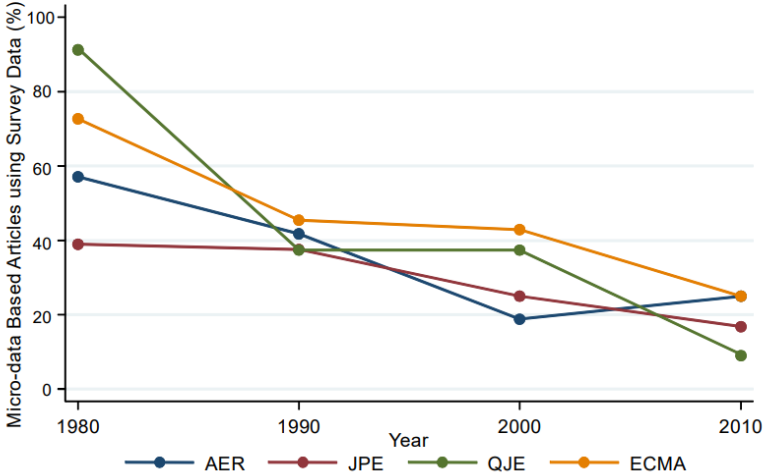
Survey vs. Register Data

Use of Administrative Data in Publications in Leading Journals, 1980-2010



Note: "Administrative" datasets refer to any dataset that was collected without directly surveying individuals (e.g., scanner data, stock prices, school district records, social security records). Sample excludes studies whose primary data source is from developing countries.

Use of Pre-Existing Survey Data in Publications in Leading Journals, 1980-2010



Note: "Pre-existing survey" datasets refer to micro surveys such as the CPS or SIPP and do not include surveys designed by researchers for their study. Sample excludes studies whose primary data source is from developing countries.

- **AMDC:** Abgestimmte Erwerbsstatistik und Registerzählung,...
- **Demographic** data: Age, gender, place of residence.
- **Education:** Degrees, training.
- **Employment** characteristics: Employment status, position in occupation.
- **Health:** Insurance status, periods of illness.
- **Taxes & Social Benefits:** Income, social assistance.
- **Family & Household:** Number of children, household size.
- **ASEP** (Austrian Socio-Economic Panel): Long-term panel survey

Why do governments implement specific measures?

- This is the question of **Political Economy**. → What influence do different policies have on individuals & the economy?
- The state tries to aggregate and weigh various interests → Democratic decision-making processes, elections.
- The state or actors within the state also have their own interests, which may go against general interests → Government failure?

Diskussion - Markt oder Staat?

Die Presse

Nachrichten

Meinung

Kardikomy

Glücksspiel: Verstaatlichung mit Tücken



WELT

HOME » WIRTSCHAFT » Coronavirus: Die Br

WIRTSCHAFT

SMART LIVING STELLENMARKT KARRIERE

WIRTSCHAFT BRITISH RAIL, BRITISH AIRWAYS

Die Briten verstaatlichen ihre Bahn – und das ist erst der Anfang

Veröffentlicht am 23.03.2020 | Lesedauer: 4 Minuten



CHRONIK ÖSTERREICH

05.04.2020

Corona-Krise: Wien bietet Firmen Teilverstaatlichung an

Die Stadt beteiligt sich bei Unternehmen, die in die Krise geschlittert sind. Insgesamt stehen 50 Millionen Euro zur Verfügung.



Angst vor Privatisierung.
Die SPÖ will Trinkwasser vor
dem Ausverkauf schützen.
Wäre ein solcher überhaupt
möglich? Der Faktcheck.

WIRTSCHAFT 10, 11

DER TAGESSPIEGEL

Telekom, Post, Bahn

Privatisierung von Staatskonzernen: Fluch oder Segen?

Der Bund ist an 108 Unternehmen beteiligt. Telekom, Post und Bahn werfen satte Gewinne ab. Dennoch sagen Kritiker: Weg mit den Anteilen. VON HEIKE JAHBERG, CARLA NEUHAUS, MARIE RÖVEKAMP, OLIVER VOSS.

HENRIK MORTSEFFER UND KEVIN P. HOFEMANN



U-AUSSCHUSS

17.12.2015

Pröll: "Verstaatlichung der Hypo war absolut richtig"

Ex-Finanzminister verteidigte Entscheidung der Hypo-Übernahme. Prölls Abrechnung zum Nachlesen.

HOME » GELD » IMMOBILIEN » Einzelhandel: Staat als Vermieter könnte Innenstädte retten

GELD

GELDLADUNG IMMOBILIEN VERBRAUCHER PODCAST BÖRSE

WELT: DEUTSCHER STADTETAG

Verstaatlichung der Innenstadt soll den Einzelhandel retten

Veröffentlicht am 02.09.2020 | Lesedauer: 4 Minuten

Von Michael Fabricius
Leitender Redakteur Immobilien



01.04.2018, 14:30 Uhr



(c) www.bilderbox.com

Discussion - Market or State?

Give your opinion!

Which goods & services should be placed **more** or **less** under the influence of the state? ...through regulations, nationalizations, taxation,

A model of Tax Indcidence



Chapter 19 (1st + Appendix)

We want to understand whether a tax increases or reduces inequality – for this, we need to find out who bears the economic burden of the tax -> **tax incidence**.

Laws determine who pays the tax. However, who actually bears the burden depends on the behavior of consumers and firms in the market – this determines the pre-tax and post-tax prices

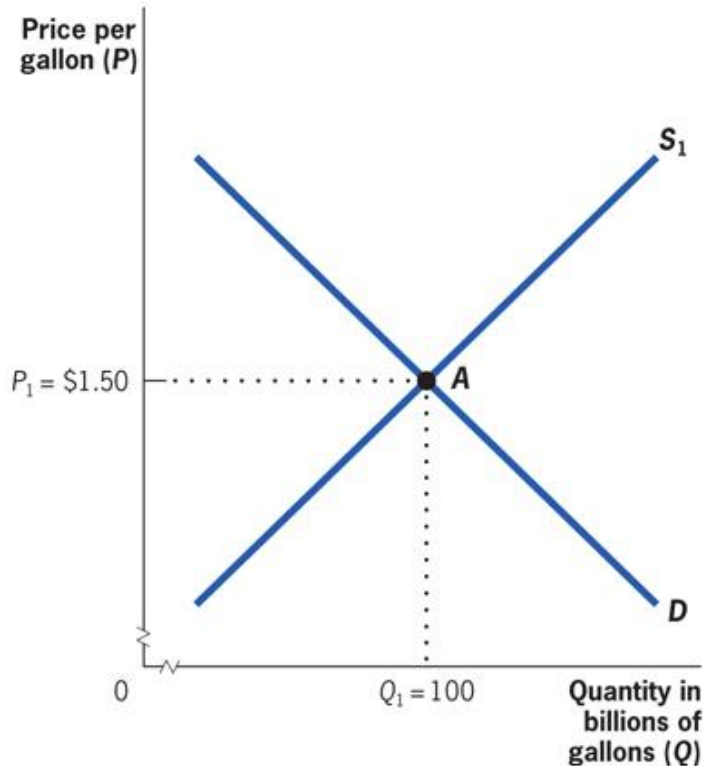
General Equilibrium: If we look beyond consumers and producers of a specific good and also analyze substitutes, complementary goods, and production factors, we are looking at incidence in a general equilibrium.

Partial Equilibrium: We, however, are only in one market (the goods market) and are looking at a specific good: we are in a partial equilibrium

Central Questions of Incidence Analysis:

- How does the market react to the introduction of a tax?
- How do prices, supply, and demand change?
- What determines behavioral changes?

Modell der Steuerinzidenz



Quelle: Gruber, 5th edition, S. 588

S = Supply curve

The quantity that producers are willing to sell at a certain price.

In perfect competition, this corresponds to the marginal costs of the producers; they sell at a price $\geq$ marginal cost.

D = Demand curve

the quantity that consumers are willing to buy at a certain price

In perfect competition, this corresponds to the willingness to pay of the consumers; they buy at a price $\leq$ willingness to pay.

A = Intersection of supply and demand
determines the market equilibrium.

Prices are always per unit (e.g., piece, hour,...).

The 3 Rules of Tax Incidence

Rule 1

Rule 1

who pays the
tax to the
government

(statutory incidence) $\neq$ (economic incidence)

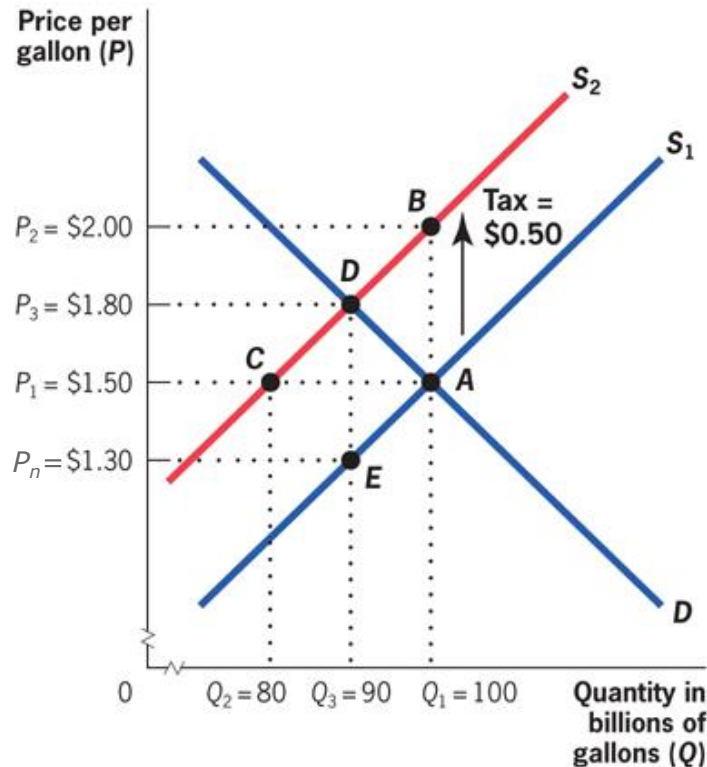
who bears the
burden

Example?

The 3 Rules of Tax Incidence

Rule 1

(a) Tax on producers



Quelle: Gruber, 5th edition, S. 588

- **Producers must pay a tax of \$0.50 per good.**
- This is equivalent to an increase in marginal costs of \$0.50.
- The supply curve shifts upwards.
- At P_1 , demand is now greater than supply ($Q_1 > Q_2$) -> Price increases to P_3 .
- New equilibrium is at D .
- Consumers must now pay a new market price of \$1.80.
- The price after tax is \$1.30 (the "real" revenue for producers).

The 3 Rules of Tax Incidence

Rule 1

The new market price ("new intersection point") = the price that is paid/received by the side that does not have to remit the tax to the state.

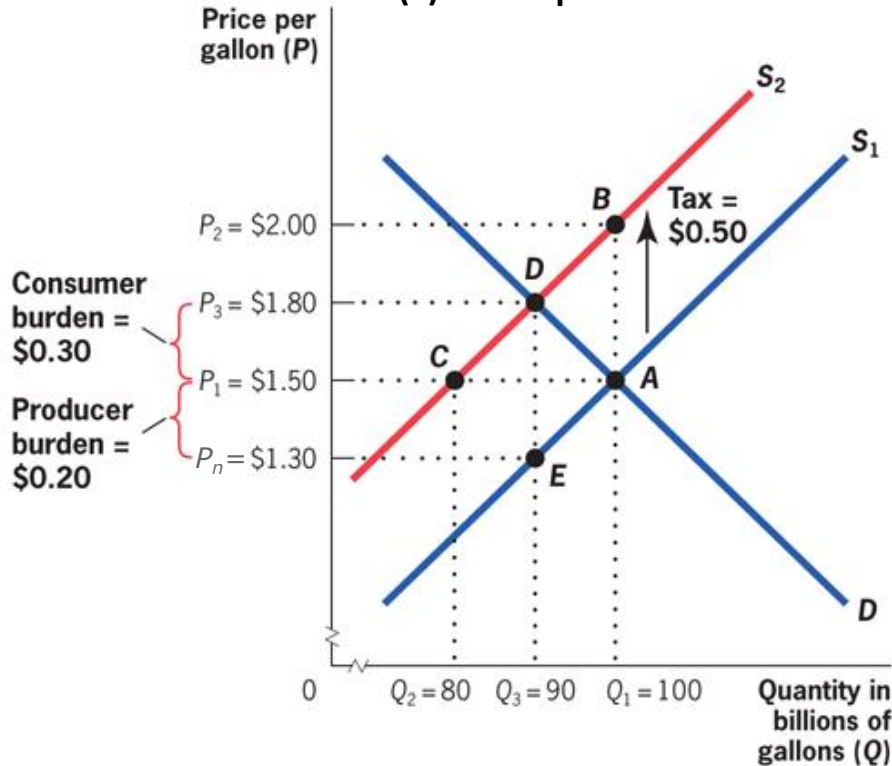
Price after tax (after-tax price) = what is paid/received by the side that must remit the tax to the state.

- **Consumers pay a tax:** Price after tax = new market price + tax
- **Producers pay a tax:** Price after tax (=revenue) = new market price - tax

The 3 Rules of Tax Incidence

Rule 1

(a) Tax on producers



Tax imposed on producers

- **New equilibrium at D:** From the \$1.80 that consumers pay, producers still have to remit a tax of \$0.50. -> Producers are left with \$1.30 (Price after tax)
- **Tax burden per unit for consumers** = $P_3 - P_1 = \$1.80 - \$1.50 = \$0.30$
- **Tax burden per unit for producers** = $P_1 - P_n = \$1.50 - \$1.30 = \$0.20$

Both sides contribute to the tax!!

Quelle: Gruber, 5th edition, S. 588

The 3 Rules of Tax Incidence

Rule 2

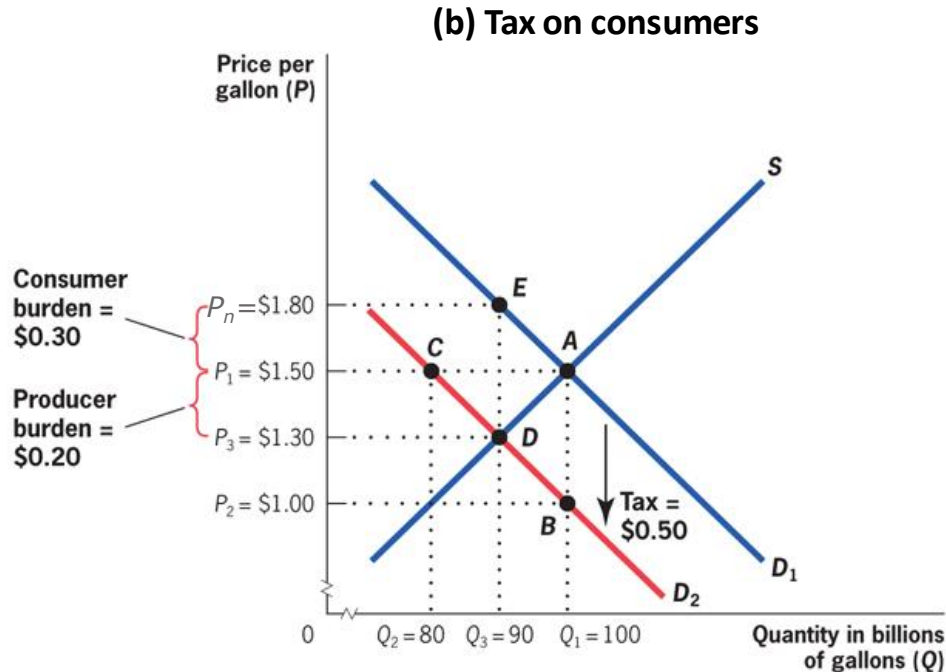
Rule 2

The distribution of the (economic) tax burden does not depend on who has to pay the tax.

The share of the tax burden borne by consumers/producers does not change if the tax is paid by consumers instead of by producers

The 3 Rules of Tax Incidence

Rule 2



Quelle: Gruber, 5th edition, S. 590

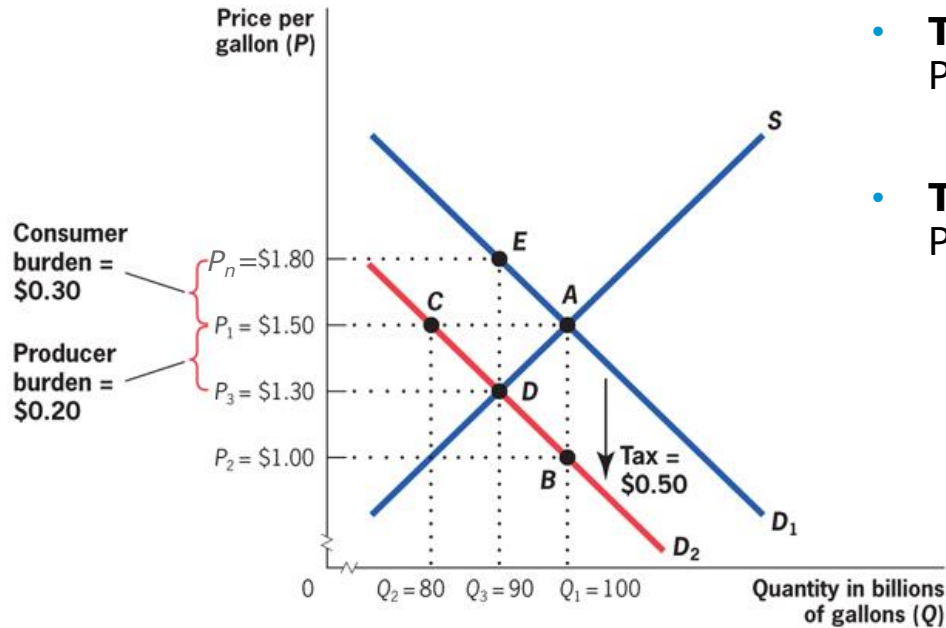
Consumers must pay a tax of \$0.50 per good.

- This is equivalent to a reduction of the Willingness to Pay by \$0.50.
- The demand curve shifts downwards.
- At P_1 , supply is now greater than demand ($Q_1 > Q_2$) -> Price falls to P_3 .
- New equilibrium is at D.
- Consumers pay \$1.80, of which \$1.30 goes to the producers and \$0.50 goes to the state.
- New market price = \$1.30
- Price after tax = \$1.80

The 3 Rules of Tax Incidence

Rule 2

(b) Tax on consumers



Quelle: Gruber, 5th edition, S. 590

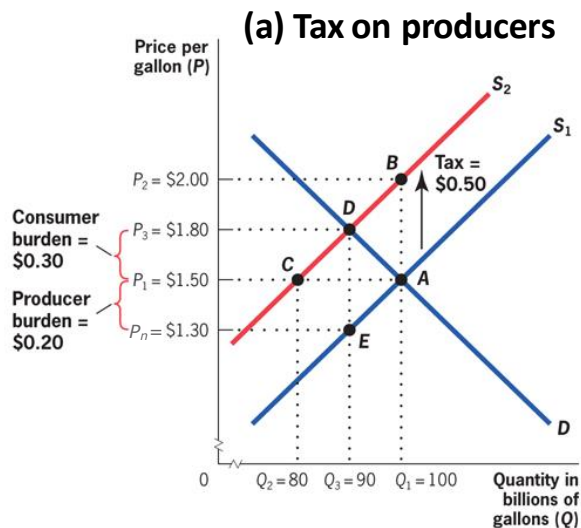
Tax imposed on consumers:

- **Tax burden per unit for consumers =**
 $P_n - P_1 = \$1.80 - 1.50 = \0.30
- **Tax burden per unit for producers =**
 $P_1 - P_3 = \$1.50 - 1.30 = \0.20

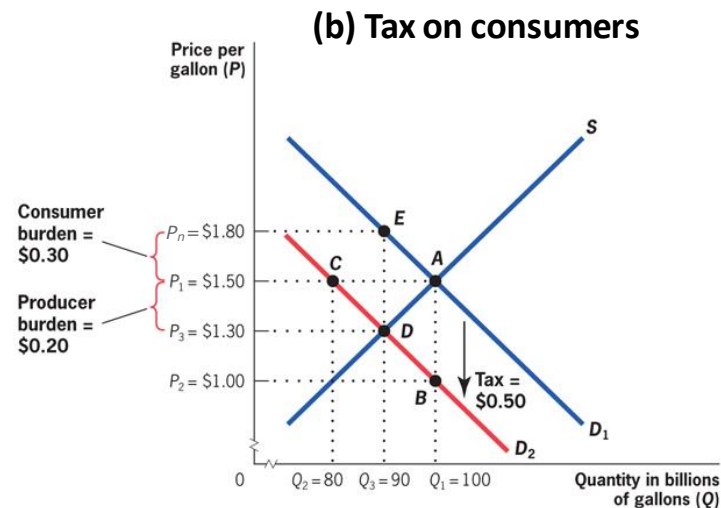
Tax burden for both sides remains the same

2. Rule at a glance

The distribution of the (economic) tax burden does not depend on who has to remit the tax.



New Market Price: \$1,80 (Tax on producers)
 Price after Tax: \$1,30
 Tax Wedge: \$0,50



New Market Price: \$1,30 (Tax on consumers)
 Price after Tax: \$1,80
 Tax Wedge: \$0,50

The 3 Rules of Tax Incidence

Rule 3

What determines who bears the economic tax burden?

How high the economic tax burden is, depends on the demand/supply elasticity.

- The more inelastic the demand/supply, the higher the burden that must be borne by consumers/producers. → The more inelastic side always bears the higher tax burden.
- That's why governments like to tax, among other reasons, alcohol, cigarettes, or gasoline.

The 3 Rules of Tax Incidence

Rule 3

Tax Shifting - "Pass-Through"

- "Tax shifting rate" or "tax pass-through" = the change in consumer price caused by the tax.
- In the long run, production occurs at constant marginal costs (due to constant returns to scale) - this corresponds to a (almost) infinitely elastic supply curve.
- This implies an almost complete shifting of the tax burden: consumption taxes are almost entirely borne by consumers (the inelastic side bears the greater part of the tax burden).
- What does the empirical evidence say? The tax shifting rate for most taxes on consumer goods is between 0.8 and 1.0.

The 3 Rules of Tax Incidence

Rule 3

Elasticity of Demand



If demand is **inelastic**, the tax is fully passed on - consumers bear the entire tax burden

If demand is **elastic**, producers bear the entire tax burden.

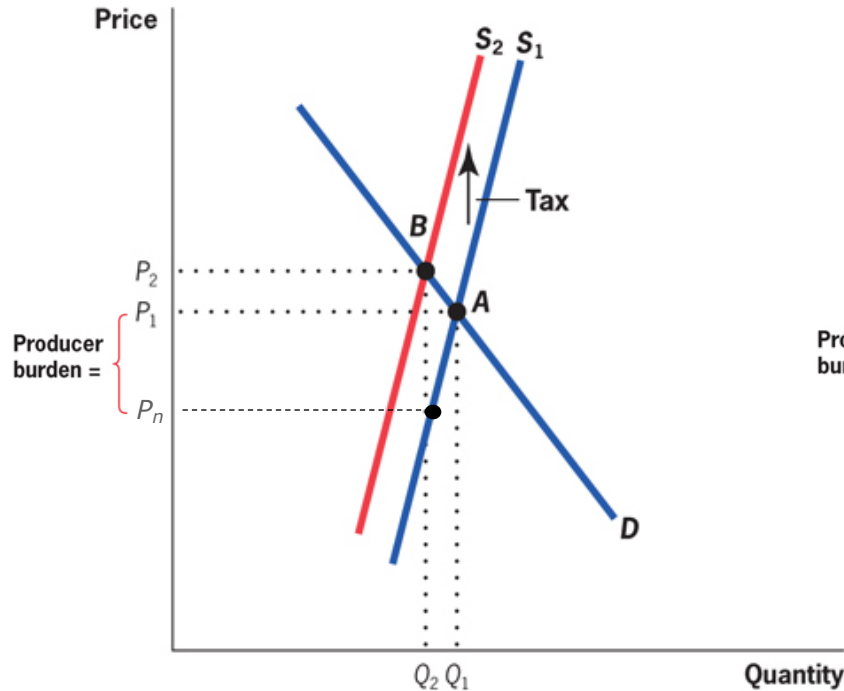
The 3 Rules of Tax Incidence

Rule 3

Elasticity of Supply

P_2 ... neuer Marktpreis:
von Konsument:innen
bezahlt

p^n ... Preis nach Steuer:
von Produzent:innen
eingekommen



If supply is **inelastic**, producers bear the higher tax burden.

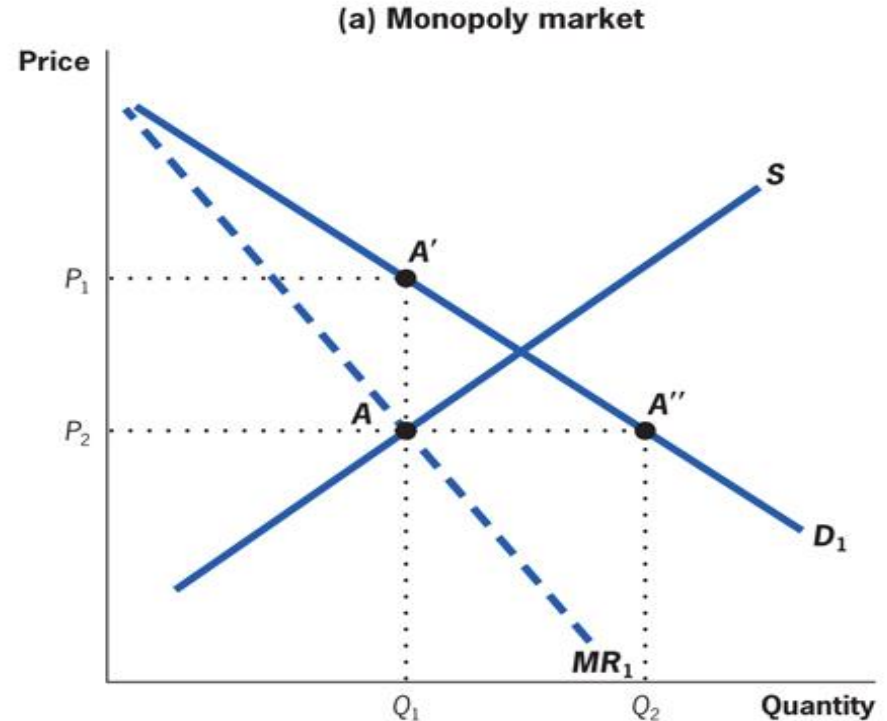


The more **elastic** the supply is, the higher the shifting of the tax burden.

The 3 Rules of Tax Incidence

in not perfectly competitive markets - Example: Monopoly market

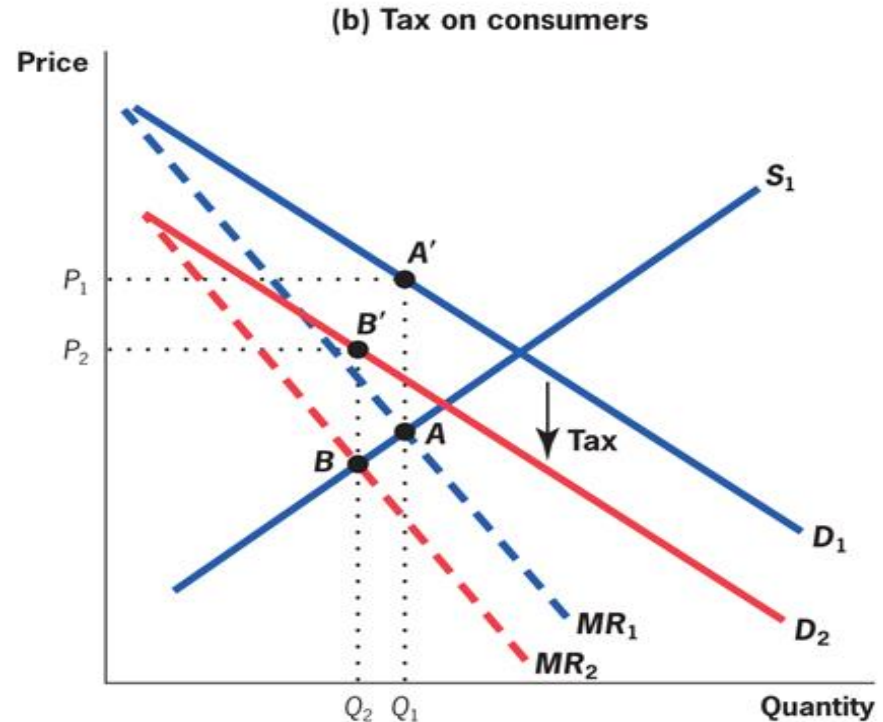
- Monopoly sets the price.
- The marginal revenue of the monopoly (MR, "how much revenue is generated by the sale of an additional unit") is below the willingness to pay of the consumers (D).
- Equilibrium condition: $MR=MC$
- Monopoly must take D into account! It cannot, for example, produce Q_1 and charge P_2 because Q_2 would be demanded at that price (A''). Instead, P_1 is charged.



The 3 Rules of Tax Incidence

in not perfectly competitive markets - Example: Monopoly market

- Example: Monopoly market
- A tax imposed on consumers shifts the demand curve and therefore also the marginal revenue $\rightarrow$ marginal revenue is dependent on the demand curve.
- $Q_1 \rightarrow Q_2$ quantity decreases, new lower price P_2 ($\rightarrow$ monopoly also bears part of the tax).
- The 3 Rules of Tax Incidence also apply to monopoly markets!



The 3 Rules of Tax Incidence

Summary

1. Tax payer $\neq$ Tax bearer
 2. The distribution of the (economic) tax burden does not depend on who has to remit the tax (= the taxpayer).
 3. The distribution of the economic burden of a tax is determined by the supply and demand elasticities.
- Important: Tax incidence refers to price changes due to the introduction of taxes (not to quantity changes -> this will come later!).

The 3 Rules of Tax Incidence

Example

The demand function in a market for a good is given by **$p = 1000 - 2x$** (x ... quantity and p ... price of the good). There is a supply monopoly for good X. (Only offered by producers)

The marginal costs of production (= supply) are **$MC = 2x$** . Fixed costs are zero.

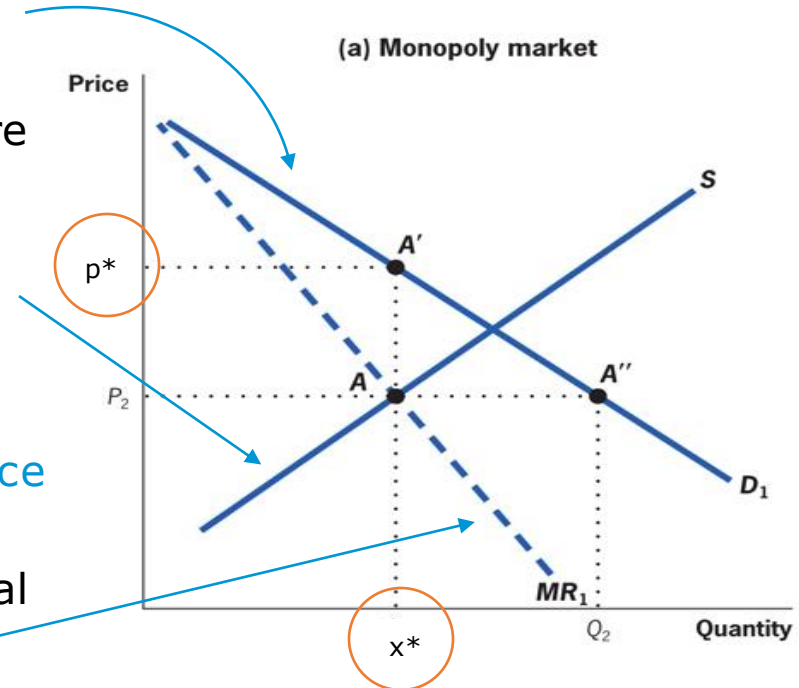
The state imposes a quantity tax with a tax amount of **$t = 30$** on the good, which must be paid by the consumers.

- a. Calculate the quantity sold and the price before the tax is introduced!
- b. Calculate the quantity sold and the price with the tax! How high is the market price (received by the producers) and the price after tax (paid by the consumers)?
- c. How is the tax burden distributed between consumers and producers?

The 3 Rules of Tax Incidence

Example

- The demand function in a market for a good is given by $p = 1000 - 2x$ (x... quantity and p... price of the good). There is a supply monopoly for good X. (Only offered by producers)
 - The marginal costs of production (= supply) are $MC = 2x$. Fixed costs are zero.
 - a. Calculate the **quantity** sold and the **price** before the tax is introduced!
- To do this, we need to set up the marginal revenue function!



Example

a. Vor Steuer:

Equilibrium condition: $MR = MC$

First, set up the revenue function,
then differentiate for marginal revenue.

$$E = p * x$$

$$E = (1000 - 2x) * x = 1000x - 2x^2$$

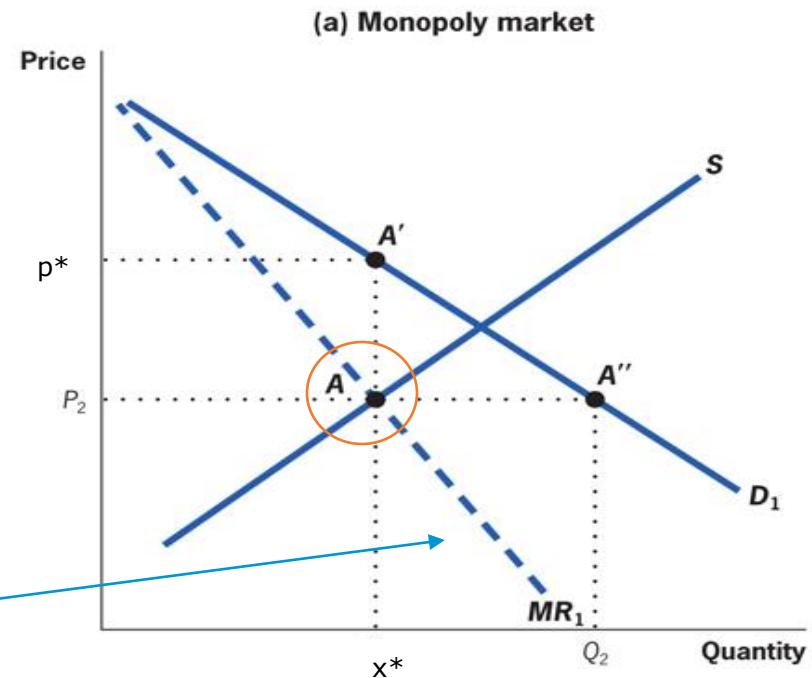
$$\rightarrow MR = 1000 - 4x$$

$$MR = MC$$

$$1000 - 4x = 2x$$

→ Solve for x , then substitute into p (D):

$$x^* = 166,67; p^* = 666,66$$

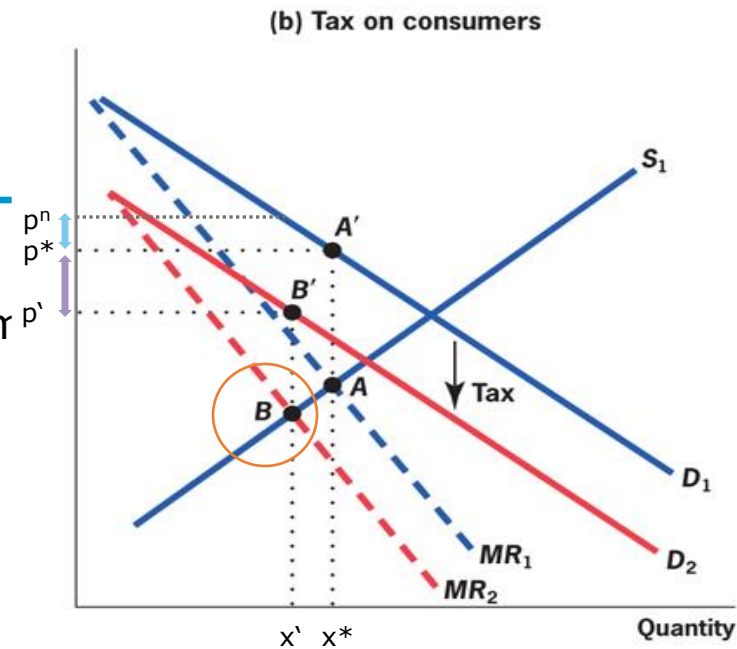


Notation: (G)E...
(Marginal) Revenue
MC... Marginal Costs t...
Tax p^* ... Equilibrium
price without tax pM ...
Market price p_n ... Price
after tax

Example

- **b. After Tax, which is paid by consumers:**
- Willingness to Pay (D) decreases by the tax, so n decreases: **$MR - t = MC$**
- $1000 - 4x - 30 = 2x$
- $\rightarrow x' = 161.67; p' = 646.66; p_n = p' + t = 676.66$

- **b. Tax Burden Distribution:**
- **Consumers:** $p_n - p^* = 676.66 - 666.66 = 10 \rightarrow 1/3$
- **Producers:** $p^* - p' = 666.66 - 646.66 = 20 \rightarrow 2/3$



Notation

(G)E... (Marginal) Revenue MC...
 Marginal Costs t ... Tax p^* ...
 Equilibrium price without tax pM ...
 Market price p_n ... Price after tax

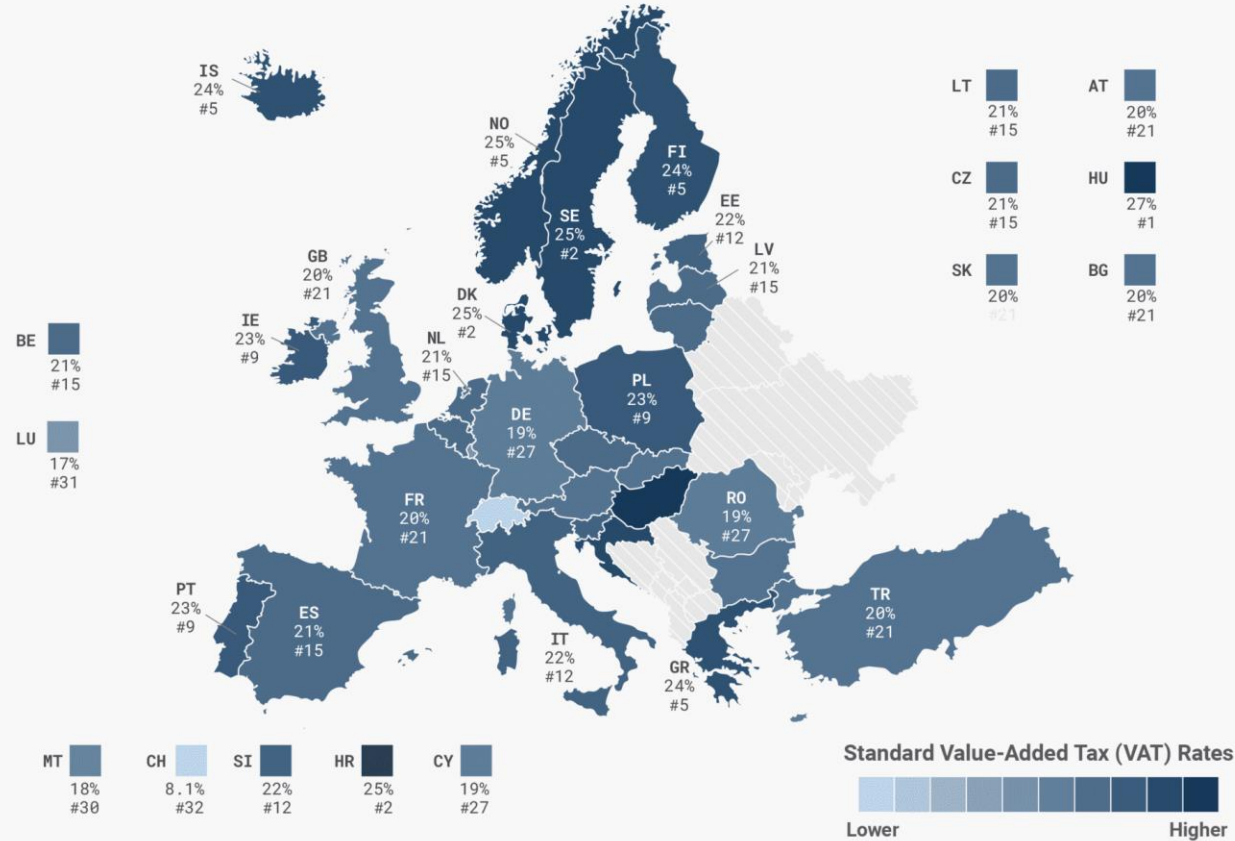
Model of Tax Incidence

critically considered: Tax Incidence Anomalies

- Youssef Benzarti (2024): "Tax Incidence Anomalies" Prepared for the Annual Review of Economics. NBER Working Paper No. 32819.
- Critically examines the three rules of tax incidence.
- To what extent do these align with empirical evidence?
- What would be needed to better model reality?
- Context: Consumption taxes (and payroll taxes in the paper).
- VAT generates the largest tax revenue worldwide.
- Almost all countries have some form of VAT (or sales tax).
- Empirical studies on consumption taxes mostly refer to VAT.

VAT Rates in Europe

Standard Value-Added Tax (VAT) Rates in European Union Member States and European OECD Countries, as of January 2024



Sources: European Commission, "Taxes in Europe Database v3," and Richard Asquith, "VAT & GST rates 2023"

Model of Tax Incidence

Critically Considered: Tax Incidence Anomalies

Rule 1

- Tax payer $\neq$ Tax bearer
- Public perception is rather that it does play a role.
- It is clearly regulated who has to pay (remit) the tax.
- But: difficult to test empirically.

The 3 Rules of Tax Incidence

Rule 1: Empirical Evidence

Kopczuk et al. (2016): "Does Tax-Collection Invariance Hold? Evasion and the Pass-Through of State Diesel Taxes". *American Economic Journal: Economic Policy*

Research Question: How does the identity of the taxpayer affect tax incidence?

- Setting: USA
- States differ in who has to remit taxes on diesel along the supply chain: retail vs. wholesale.
- The study evaluates changes in this structure.

The 3 Rules of Tax Incidence

Rule 1: Empirical Evidence

- **Results**
- It does matter who has to remit the tax.
- If the retailer remits the tax, the pass-through of the fuel tax to the diesel price is lower.
- Shifting the tax collection to the wholesaler i) increases the pass-through of the diesel tax to consumers and ii) tax revenues increase.
- Why? Hypothesis: Opportunities for tax evasion differ along the supply chain; tax evasion is easier at the retail level (e.g., through false declaration of quantities or use of diesel).
- -> Interesting implications for CO2 taxation and where it should be applied

The 3 Rules of Tax Incidence

Rule 1: Empirical Evidence

- Chetty, Looney, Kroft (2009) *Salience and Taxation: Theory and Evidence*
- **Research Question:** What role does the salience ("conspicuousness") of taxes play in consumer decisions?
- Field experiment in a supermarket in the USA: Prices for a subgroup of hygiene articles were displayed inclusive of value-added tax (This is normally not the case in the USA).
- **Results**
- The demand for hygiene articles labeled with value-added tax decreases!



Source: Chetty, Looney, and Kroft (2009)

The 3 Rules of Tax Incidence

Rule 1: Empirical Evidence

2. Comparison of a tax increase on alcohol at two different tax rates in the USA.

- Excise taxes are included in the price -> "price increase"
- VAT is not included in the price -> "tax increase"
- According to the rules of tax incidence, the effect should be the same (regardless of who pays the tax).

Results

- Demand changes more strongly (4x) with changes in excise tax (which is shown on the price tag) than with VAT.
- Given a lack of reaction to a VAT increase, there is less pressure on firms to adjust their prices when the tax is increased.
- It is not ignorance about taxes that is decisive, but salience!

Taxes on producers are fundamentally "visible," like the excise taxes here.

- -> **Relevant on which side of the market the tax is collected!**

- **Rule 2**
- Does the distribution of the (economic) tax burden depend on who has to remit the tax (= the taxpayer)?
- Example of tax cuts: had many goals already
 - Boosting demand (e.g., after recessions)
 - Idea: Incidence is borne by consumers, i.e., prices should fall.
 - Support for businesses (e.g., hospitality industry during/after pandemic)
 - Idea: Incidence is borne by producers, i.e., prices should remain similar.
 - Mitigating the effects of inflation.
- → different goals through one measure, but: If the tax burden is always distributed equally, regardless of who pays, how is that possible?

The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

"Who Really Benefits From Consumption Tax Cuts? Evidence From a Large VAT Cut in France" Youssef Benzarti and Dorian Carloni. AEJ: Economic Policy, 2019.

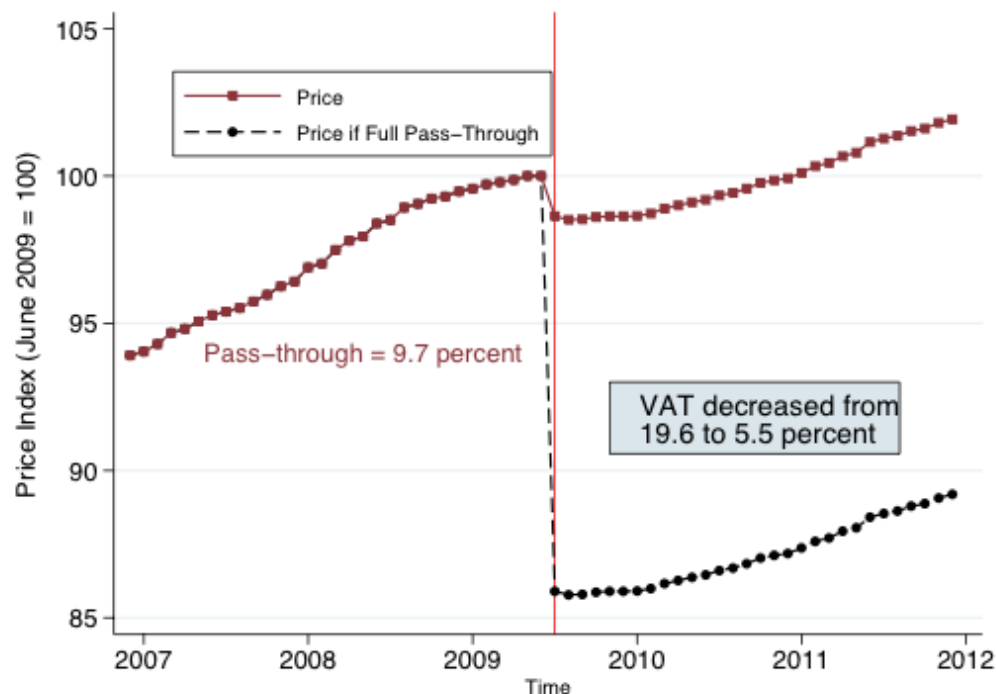
- **Reform:** In 2009, France's value-added tax for meals in restaurants was lowered from 19.6% to 5.5%.
- **Goal:** The tax cut should be equally shared among restaurants, employees, and consumers.
- **Setup**
 - To determine who effectively benefits from this tax cut, national price data and company balance sheets are used.
 - Analysis of the distribution of the tax burden between consumers and producers & how the burden is distributed among the individual factors of production.

Short and clear summary of the paper → [hier](#)

The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

Figure 1 Restaurant meal prices around the July 2009 VAT reform



- While the tax is reduced by 14.1 percentage points, prices only fall by 1.36%.
- Pass-through: $\frac{1.36}{14.1} = 9.7\%$
 - It is mainly the restaurants that benefit
- The quantity sold also does not change.

Notes: This figure plots the evolution of the prices of sit-down restaurants around the July 2009 reform and compares it to a hypothetical counterfactual had there been 100% pass-through of the VAT reduction.

The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

- How is the relief distributed within the individual factors of production?
 - Employees?
 - Restaurant owners?
 - Food suppliers?
 - Investors?
-
- Public discourse often assumes that most of the incidence lies with consumers - but studies show that firms usually benefit.

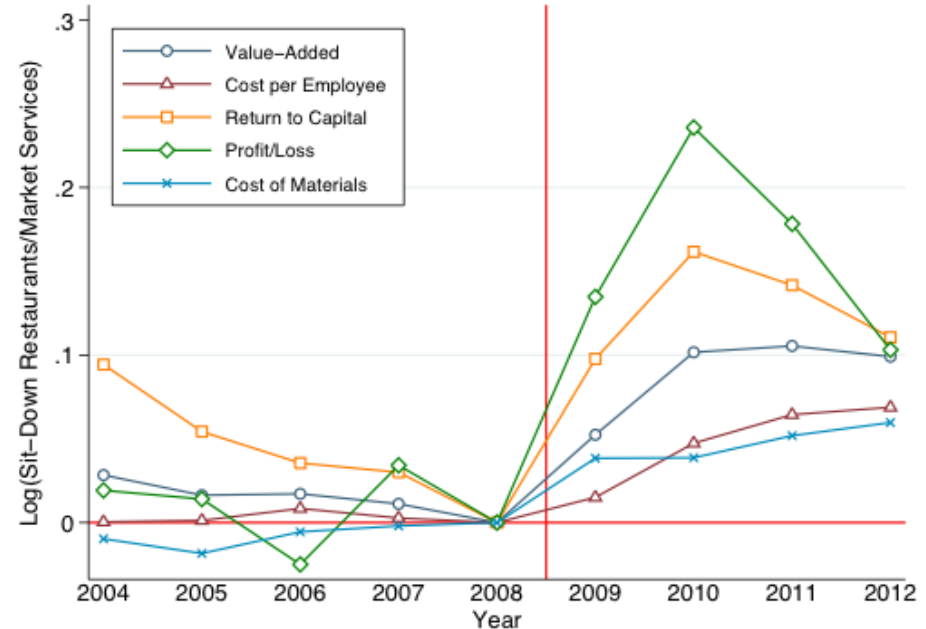
The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

Figure 8: Estimated Incidence of the VAT Reform



Effekt der Steuersenkung im Vergleich zu Kontrollgruppe:



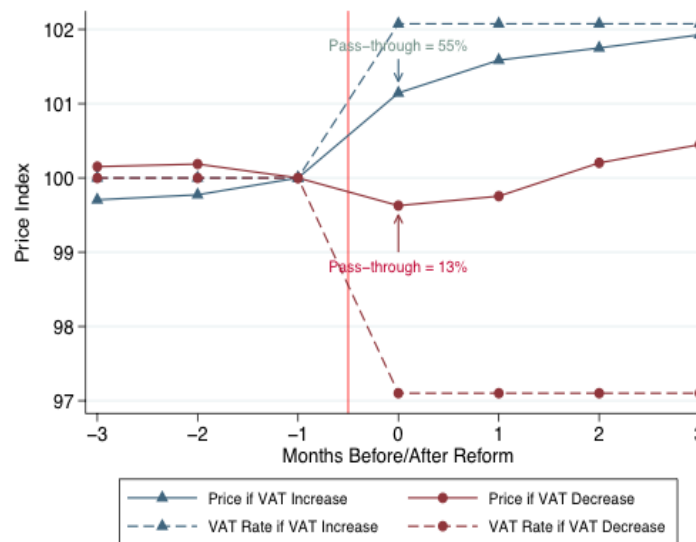
The restaurant owners and investors benefit the most. Employment did not increase (not even in the long term). In the longer term, employees also benefit through higher wages.

The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

- The pass-through of tax increases to consumers is 4-5 times as high as that of tax cuts.
- Temporary tax cuts in times of crisis can lead to long-term higher prices.

Figure 4 Aggregated VAT changes in Europe, 1996–2015



Notes: This figure aggregates all VAT changes that occurred in Europe from 1996 to 2015, and plots the response of prices to VAT increases versus VAT decreases. It shows that the pass-through of prices is larger for VAT increases than decreases.

The 3 Rules of Tax Incidence

Rule 2: Empirical Evidence

- **Temporary VAT reduction Austria Covid-19**
- Covid-19: **Reduction of the VAT by up to 5%** in accommodation, tourism, gastronomy, culture, and media in Austria to compensate for the collapse in consumption.
- Here too, there was hardly any pass-through -> mainly helped businesses.
- Most positive effects for high-turnover, large businesses (often not so badly affected by Covid-19, e.g., McDonald's).
- [WIFO-Studie](#)

Model of Tax Incidence

Critically Considered: Tax Incidence Anomalies

Rule 3

- Is the distribution of the economic burden of a tax determined by the supply and demand elasticities?

The 3 Rules of Tax Incidence

Rule 3: Empirical Evidence

Marion and Muehlegger (2011) "Fuel Tax Incidence and Supply Conditions"
Journal of Public Economics

- Investigates the tax incidence of taxes on diesel and gasoline in the USA from 1983-2003 (panel data, monthly at the state level).
- Additionally, it captures factors that can influence the elasticity of supply (and thus the distribution of the tax burden):
 - Capacity constraints
 - Environmental regulations

Results

- Normally, there is **full pass-through** (or more) of taxes to consumers.
- In periods when there are capacity constraints on the producers' side (supply is less elastic), there is less pass-through. -> Producers bear part of the tax burden.

The 3 Rules of Tax Incidence

Rule 3: Empirical Evidence

- **Theoretical Implications:**

- Capacity constraints -> Elasticity of supply decreases -> Producers bear more of the tax burden.

- **Policy Implications:**

- Since demand for diesel and gasoline is inelastic and there is full pass-through in normal times -> No compensation for producers is necessary.
- In times of high demand (and high prices) or other supply-side bottlenecks, "tax holidays" do not only benefit consumers, as producers also bear part of the tax burden in these cases.

The 3 Rules of Tax Incidence

Rule 3: Empirical Evidence

Harju et al (2018): "Firm types, price-setting strategies, and consumption-tax incidence". *Journal of Public Economics*

- Tax cut for restaurants in Finland.
- Restaurant chains: first pass on 100% of the tax cut and then decrease the pass-through over time.
- Individual restaurants: first pass on 0% and remain at about that level.

Benzarti et al. (2024): "Can VAT cuts and anti profiteering measures dampen the effects of food price inflation?" NBER Working Paper

- Reduction of VAT on food in Argentina.
- Supermarket chains pass on about 80% of a tax cut.
- Individual supermarkets pass on about 30%, approaching 0% over time.
- No convergence over time.
- -> The company structure plays a role in the distribution of the tax burden.

Model of Tax Incidence

Critically Considered: Tax Incidence Anomalies

- What can we still learn from the tax incidence model?
- It is a popular model because it is simple, has a clear connection to empirical parameters (elasticities), and can estimate tax incidence.
- Many statements are derived from the model -> important to know where they come from.
- But the evidence shows: Tax incidence varies greatly depending on the context! What could we add to the model?
- Different reactions to tax increases and decreases.
- An indicator for the visibility of the tax.
- Degree of tax evasion.
- Firm structure.
- In practice, this means estimating a lot more elasticities - depending on the context!